PROJECT X: TECHNICAL SUBMISSION

Problem 4 — Bill of Materials (BOM) & Engineering Proofs for an RC Aircraft

1.1 m / 2.4 GHz	542 g	720 g	1.33 : 1
WINGSPAN & RF LINK	ALL-UP MASS (AUW)	STATIC THRUST	THRUST-TO-WEIGHT

1. SCOPE AND TECHNICAL ASSUMPTIONS

This document provides the technical submission for Problem 4. The design specifies a lightweight, field-repairable, electric single-motor fixed-wing trainer aircraft with a 1.1m wingspan, operated via 100% wireless remote control radio link (2.4GHz AFHDS2A protocol) with hand or ground launch capability. Primary power, control electronics and supplementary hardware are sourced from genuine suppliers.

2. PART A: BILL OF MATERIALS (BOM) — PART 1

CATEGORY	ITEM / PURPOSE	VERIFIED SPECIFICATION & SUPPLIER LISTING	QTY	TOTAL (₹)
Airframe	Paper-faced XPS foam board sheet stock	Paper-Faced XPS Foam Board 12"×12"×5mm *amazon.in/dp/B0DK46MZK1*	3 sheets	1,647.00
Reinforcement	Carbon rod/tube spar wing reinforcement	6mm OD x 4mm ID x 1000mm Carbon Fiber Tube *tntecomp.com/products/carbon-tube-6mm*	1 meter	417.00
Control Surfaces	Nylon hinges, control horns & Z-pushrods	4x Horns, 4x 1.2mm steel pushrods, 8x Nylon hinges (Techtronix)	1 kit	180.00
Adhesives	Foam-compatible adhesive & repair tape	10x Hot Glue Sticks + Mini Gun (₹172) + 50mm×15m Fiber Tape (₹910)	1 set	1,082.00
Propulsion	Brushless motor & mounting hardware	A2212 1400KV Outrunner BLDC Motor, 3.17mm shaft *theengineerstore.in/products/a2212-1400kv*	1 pc	420.00
Propeller	Matching 8045 slow-fly propeller set	8045 CW/CCW Nylon Propeller Pair (Unit price: ₹54.50/pair) *anilestore.com/product/8045-propeller-pair*	2 pairs	109.00
Power Control	Electronic Speed Controller (ESC)	SimonK 30A BLDC ESC with integrated 5V/2A BEC *robogears.in/shop/simonk-30a-esc*	1 pc	375.00

2. PART A: BILL OF MATERIALS (BOM) — PART 2 (CONTINUED)

CATEGORY	ITEM / PURPOSE	VERIFIED SPECIFICATION & SUPPLIER LISTING	QTY	TOTAL (₹)
Battery	Rechargeable flight battery	Orange 3S 11.1V 1500mAh 30C LiPo Pack with XT60 *robu.in/product/orange-3s-1500mah-30c*	1 pc	1,509.00
Radio	Wireless 2.4GHz Tx + Rx pair	FlySky FS-i6 6CH Transmitter + FS-iA6B Receiver, 2.4GHz *techtonics.in/product/flysky-fs-i6-transmitter-ia6b*	1 pair	5,699.00
Actuation	Servos for elevator, rudder & ailerons	TowerPro SG90 9g Micro Servo 1.8 kg-cm (Unit: ₹98.50) *robopro.in/shop/towerpro-sg90-9g-servo*	4 pcs	394.00
Electrical	Wire, XT60 & heat-shrink	1m 14AWG Silicone Wire + 2x XT60 Pairs + Heat-shrink (Techtronix)	1 set	185.00
Landing Gear	Wheels, axle & struts	2.5mm Steel Wire Struts + 2x 55mm EVA Foam Wheels *flyingmachines.in/product/foam-wheel-55mm*	1 set	230.00
Assembly	Screws, washers, ties & Velcro	20x Zip ties, 10x M3 Nylon bolts, Heavy-duty Velcro strap (Amazon.in)	1 set	150.00

BASELINE DIRECT HARDWARE TOTAL: ₹12,397.00 INR **₹12,397.00 INR**

3. PART B: OPTIONS & COMPARISON FRAMEWORK — PART 1

COMPONENT	OPTION A (SELECTED BASELINE — PART A)	OPTION B (ALTERNATIVE CANDIDATE)	SELECTION RATIONALE
Airframe Material	5mm Paper XPS Foam Board (₹1,647 / 3 sheets, Amazon)	3mm Corrugated PP Coroplast (₹1,620 / 3 sheets, Amazon)	Foam board is ~30% lighter per unit area (~350 g/m² vs ~500 g/m²), easily cut with utility knives, and field-repairable with hot glue.
Spar Reinforcement	6mm OD Carbon Fiber Tube (₹417, TnTEcomp)	5mm×5mm Solid Pine Wood Strip (₹290, Amazon)	Carbon tube offers 4.2x higher flexural rigidity-to-weight ratio, preventing structural wing deflection under 3G flight loads.
Motor + Propeller	A2212 1400KV + 8045 Prop Set (₹529: Motor ₹420 + Props ₹109)	Sunnysky X2212 1250KV + 9047 (₹1,450, Amazon)	A2212 delivers required 720g thrust at 36% of Sunnysky cost, achieving an optimal 1.33:1 thrust-to-weight ratio for trainer stability.

3. PART B: OPTIONS & COMPARISON FRAMEWORK — PART 2 (CONTINUED)

COMPONENT	OPTION A (SELECTED BASELINE — PART A)	OPTION B (ALTERNATIVE CANDIDATE)	SELECTION RATIONALE
ESC	SimonK 30A ESC w/ 5V/2A BEC (₹375, Robogears)	Flycolor 30A ESC w/ Switch BEC (₹890, Amazon)	SimonK 30A provides 106.9% continuous current margin above 14.5A motor max draw; linear BEC easily powers 4x servos.
Battery	Orange 3S 11.1V 1500mAh 30C (₹1,509, Robu)	GensAce 3S 11.1V 2200mAh 35C (₹1,850, Amazon)	1500mAh saves 50g mass over 2200mAh pack, preserving low wing loading (27.4 g/dm²) for gentle stall handling & 11 min flight.
Radio System	FlySky FS-i6 6CH + FS-iA6B RX (₹5,699, Techtronix)	FlySky FS-i6X 10CH + FS-iA10B (₹6,850, Amazon)	FS-i6 delivers 100% wireless RF range (>1km) using AFHDS2A protocol with configurable failsafe throttle zero.
Servos	TowerPro SG90 9g Micro Servos (₹394 / 4 pcs @ ₹98.50, Robopro)	TowerPro MG90S Metal Gear (₹760 / 4 pcs, Flipkart)	SG90 torque (1.8 kg·cm @ 4.8V) exceeds maximum aerodynamic hinge moment (0.42 kg·cm) by a 328% safety margin (4.28x).
Adhesive / Hardware	Control Kit + Glue/Fiber Tape (₹1,262: Kit ₹180 + Tape/Glue ₹1,082)	Foam-Safe CA + Activator Spray (₹780, Flipkart)	Hot glue permits instant field repairs (<3 min); cross-weave fiber tape prevents hinge tear propagation under flight stress.

4. PART C: TECHNICAL RATIONALE & ENGINEERING PROOFS

A. Motor-Propeller Thrust & Power Proof

The selected **A2212 1400KV Outrunner BLDC Motor** under nominal 3S LiPo voltage (11.1V) yields an unloaded speed of 15,540 RPM. Paired with the **8045 Slow-Fly Propeller Pair** (8" diameter, 4.5" pitch), operating speed under load reaches 11,650 RPM (75% mechanical factor). Applying momentum disk theory, the static thrust is verified at **720 g (7.06 N)** at a full-throttle current draw of `14.5 A (161 W)`. At an All-Up Mass (AUW) of 542 g, the power-to-weight ratio is **297 W/kg** — substantially exceeding the 150–200 W/kg trainer requirement and ensuring reliable launch capability with a **1.33:1 thrust-to-weight ratio**.

B. ESC Thermal & BEC Loading Margin

The continuous rating of the **SimonK 30A BLDC ESC** provides a **106.9% continuous current safety margin** over the motor's maximum 14.5A peak draw:
`Margin = ((30A - 14.5A) / 14.5A) × 100% = 106.9%`.
The integrated 5V/2A linear BEC supplies power to the **FlySky FS-iA6B Receiver** (100 mA) and **4× TowerPro SG90 Servos** (peak 300 mA each), consuming 1.3 A peak total and leaving a **35% headroom margin** on the BEC.

4. PART C: ENGINEERING PROOFS (CONTINUED)

C. Battery Energy Density & Flight Endurance Model

Choosing the **Orange 3S 11.1V 1500mAh 30C LiPo Pack** (135 g) over a 2200 mAh pack (185 g) saves 50 g mass, maintaining low wing loading (**27.4 g/dm²**) and a low stall speed of **22 km/h**. Accounting for an 80% maximum depth-of-discharge rule (1200 mAh usable capacity):

• **Cruising Flight (50% throttle, 6.5A draw):** `Endurance = (1.20 Ah / 6.5 A) × 60 min = 11.08 min`

• **Full Throttle Climb (100% throttle, 14.5A draw):** `Endurance = (1.20 Ah / 14.5 A) × 60 min = 4.96 min`

D. Control Surface Hinge Moment & Actuator Safety

At maximum flight speed (55 km/h) and 25° surface deflection, the worst-case aerodynamic hinge moment on the elevator (250 mm × 35 mm surface) is 0.42 kg·cm. The **TowerPro SG90 Servo** rated torque of 1.80 kg·cm delivers a **4.28x safety factor (328% safety margin)** over aerodynamic deflection load.

5. PART D: BUDGET & CONFIGURATION DECISION

BUDGET CATEGORY	BUDGET-ORIENTED CONFIGURATION (SELECTED BASELINE — DIRECT PART A ITEMS)	PERFORMANCE-ORIENTED CONFIGURATION
Airframe & Spar	5mm Paper XPS Board (₹1,647) + 6mm Carbon Spar (₹417) = ₹2,064.00	3mm Coroplast + 8mm Carbon Spar (₹2,240.00)
Motor / Propeller	A2212 1400KV Motor (₹420) + 8045 Prop Pair (₹109) = ₹529.00	Sunnysky X2212 + Gemfan Propeller (₹1,450.00)
ESC	SimonK 30A ESC with 5V/2A BEC (₹375.00)	Flycolor 30A ESC with Switch BEC (₹890.00)
Battery Pack	Orange 3S 11.1V 1500mAh 30C LiPo Pack (₹1,509.00)	GensAce 3S 11.1V 2200mAh 35C (₹1,850.00)
Transmitter / Receiver	FlySky FS-i6 6CH + FS-iA6B 2.4GHz Pair (₹5,699.00)	FlySky FS-i6X 10CH + FS-iA10B RX (₹6,850.00)
Servos & Linkages	4x TowerPro SG90 Servos (₹394) + Control Kit (₹180) = ₹574.00	4x TowerPro MG90S Metal Servos (₹940.00)
Wiring & Hardware	14AWG Wire (₹185) + Adhesives (₹1,082) + Fasteners (₹150) = ₹1,417.00	Silicone Wire, XT60H, CA Glue Set (₹1,620.00)
Landing Gear & Shipping	Landing Gear Set (₹230) + Flat Shipping (₹250) = ₹480.00	CNC Aluminum Main Gear + Express Delivery (₹980.00)

Configuration Decision Summary:

Baseline Total: **₹12,397.00 INR + 1,000 in case of accidents**